Top 5 open source AI projects for beginners

Teachable Machine (Google)

A no-code AI platform that simplifies image, audio, and pose recognition. This tool is ideal for beginners as it enables them to build AI models without writing extensive code.

Mini project: Create an image classifier for hand gestures using Teachable Machine
Pseudo-code:
Step 1: Visit *https://teachablemachine.withgoogle.com/*
Step 2: Upload hand gesture images (e.g., thumbs up, thumbs down).
Step 3: Train the model using Teachable Machine's interface.
Step 4: Export the model and integrate it into a web or mobile application.

Hugging Face Transformers

Provides pre-trained NLP models for tasks like sentiment analysis and text classification. Beginners can use these models without deep AI expertise.

Mini project: Analyse sentiment from Twitter data using a pre-trained model
Pseudo-code:

```
from transformers import pipeline
# Load pre-trained sentiment analysis model
sentiment_pipeline = pipeline("sentiment-analysis")
# Analyze sentiment of a tweet
tweet = "I love open-source AI!"
result = sentiment_pipeline(tweet)
print(result)
```

OpenCV for computer vision

A popular library for image and video processing, widely used in AI-driven applications.

Mini project: Face detection using OpenCV and Python
Pseudo-code:

```
import cv2
# Load the pre-trained Haar Cascade face detector
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades +
"haarcascade_frontalface_default.xml")
# Read image
img = cv2.imread("face.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
# Detect faces
faces = face_cascade.detectMultiScale(gray, 1.1, 4)

# Draw rectangles around detected faces
for (x, y, w, h) in faces:
    cv2.rectangle(img, (x, y), (x+w, y+h), (255, 0, 0), 2)
cv2.imshow("Face Detection", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

FastAI for deep learning

Simplifies neural network training with pre-built modules, making it easier for beginners to train AI models with minimal coding.

Mini project: Train an image classifier in a few lines of code
Pseudo-code:

```
from fastai.vision.all import *
# Load dataset
path = untar_data(URLs.PETS)/'images'
dls = ImageDataLoaders.from_name_func(path, get_image_
files(path), lambda x: x[0].isupper(), item_tfms=Resize(224))
# Train model
learn = cnn_learner(dls, resnet34, metrics=accuracy)
learn.fine_tune(1)
```

Ludwig by Uber (No-code AI)

Enables AI model training without requiring extensive programming knowledge; perfect for beginners looking to explore AI without deep technical expertise.

Mini project: Predict customer churn using structured data
Pseudo-code:

```
from ludwig.api import LudwigModel
import pandas as pd

# Load dataset
data = pd.read_csv("customer_churn.csv")
# Define Ludwig configuration
yaml_config = {
    "input_features": [{"name": "customer_type", "type":
"category"}],
    "output_features": [{"name": "churn", "type": "binary"}]
}
# Train model
model = LudwigModel(config=yaml_config)
train_stats = model.train(dataset=data)
```

Real-life AI applications using easy tools

AI-powered chatbot with Rasa

Rasa is an open source conversational AI framework that allows developers to build intelligent chatbots. It supports